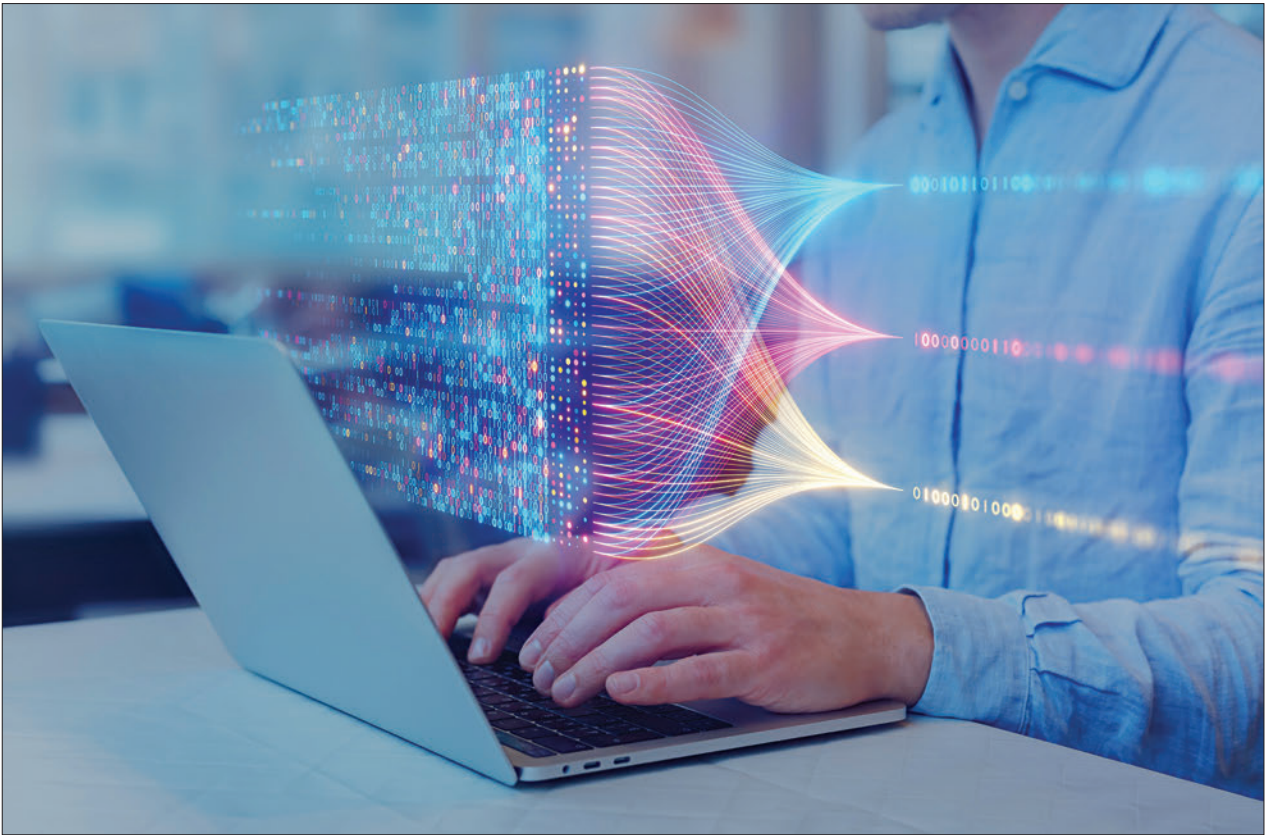




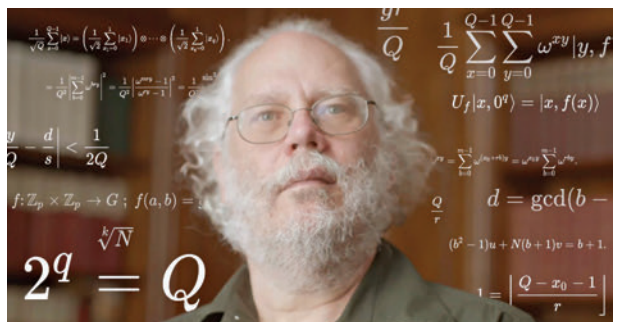
Quantum Algorithms

Unlocking quantum potential, from prime factorization to machine learning innovations

Quantum algorithms represent the pinnacle of quantum computing, offering solutions to problems once deemed infeasible for classical computation. This part of Chapter 4 delves into two groundbreaking algorithms: Shor's Algorithm, which revolutionised the field of cryptography, and Grover's Algorithm, which presents a superior approach to searching unsorted databases.

Shor's Algorithm

Imagine effortlessly cracking the codes that safeguard our online transactions, confidential communications, and even military secrets. This isn't science fiction, but a potential reality posed by Shor's algorithm, a quantum computing innovation that threatens to dethrone the kings of encryption.



Developed in 1994 by Peter Shor, this algorithm tackles the seemingly simple task of factoring large numbers—breaking them down to their prime components. While this might seem